215316: metasandstone, Midlands Road

(Noondine Chert, Coomberdale Subgroup, Moora Group, Moora Basin, Pinjarra Orogen)

Location and sampling

Moora (SH 50-10), Moora (2136)
MGA Zone 50, 409495E 6620582N

Sampled on 11 August 2016

This sample was collected from an outcrop of mixed sandstone and chert, about 11.5 km north-northeast of Moora, 3.3 km southeast of Coomberdale mine, and 2.9 km east of the Midlands Road.

Tectonic unit/relations

The unit sampled is the Noondine Chert of the Moora Basin. The Noondine Chert is the youngest stratigraphic component of the Coomberdale Subgroup of the Moora Group (Myers, 1990). The Moora Group unconformably overlies Archean rocks of the Yilgarn Craton immediately to the east of the Darling Fault, and is divided into the basal Billeranga Subgroup and the disconformably overlying Coomberdale Subgroup (Myers, 1990). The Moora Group is intruded and locally contact-metamorphosed by north-trending dolerite dykes of the c. 1210 Ma Boyagin Dolerite (Wingate, 2007). The Coomberdale Subgroup is up to 1500 m thick and comprises sandstone, siltstone, dolomite, and silicified dolomite (Carter and Lipple, 1982). The Noondine Chert crops out as a series of low hills to the east of the Perth Basin extending south from near the town of Three Springs to Moora, a distance of about 150 km. The unit consists of a sequence of chert, quartzite, chert breccia and partly silicified (commonly dolomitic) limestone and dolomite (Playford, 1976; Carter and Lipple, 1982; Abeysinghe, 2003). This sample was collected to determine a maximum depositional age and provenance for the Noondine Chert.

Petrographic description

The sample is a coarse-grained quartz metasandstone (Fig. 1), consisting of about 75–80% detrital quartz and 20–25% microcrystalline siliceous material, and accessory zircon. Detrital quartz grains are up to 2 mm in size and average 0.7 to 0.8 mm. Most grains are subrounded, although some are well-rounded and a few of the smallest grains are subangular. Most grains are moderately strained and there is a small amount of subgraining but minimal recrystallization. Packing of the detrital quartz grains varies at thin-section scale from tightly packed, displaying slight development of sutured boundaries, to a more open framework. The microcrystalline material that occurs between the quartz grains is mainly cherty and siliceous in appearance although in places includes patches of sericite. Some of the sericitized material includes transparent, probably detrital, grain outlines that are visible in plane-polarized light, although some forms thin veneers around detrital quartz and may represent matrix. Assuming most material was originally detrital, the textural maturity of the sandstone is probably submature or mature. The rock has been weakly deformed at low metamorphic grade.

Zircon morphology

Zircons isolated from this sample are colourless to dark brown or opaque, anhedral to subhedral, and variably rounded. The crystals are up to 300 µm long, and equant to elongate, with aspect ratios up to 5:1. Many crystals have pitted outer surfaces and, in cathodoluminescence (CL) images, concentric zoning truncated at grain boundaries, features consistent with sedimentary transport. A CL image of representative zircons is shown in Figure 2.

Analytical details

This sample was analysed on 19–20 and 22–24 October 2018, and 4–5 November 2021, using SHRIMP-B. Analyses 1.1 to 21.1 (spot numbers 1–21) were obtained during the first session, together with 10 analyses of the M257 standard, of which five analyses indicated an external spot-to-spot (reproducibility) uncertainty of 0.50% (1σ) and a $^{238}U/^{206}Pb^*$ calibration uncertainty of 0.28% (1σ). Analyses 76.1 to 75.1 (spot numbers 22–77) were obtained during the second session, together with 25 analyses of the M257 standard, of which 23 analyses indicated an external spot-to-spot (reproducibility) uncertainty of 1.19% (1σ) and a $^{238}U/^{206}Pb^*$ calibration uncertainty of 0.32% (1σ). Analyses 116.1 to 115.1 (spot numbers 78–117) were obtained during the third session, together with eight analyses of the M257 standard, which indicated an external spot-to-spot (reproducibility) uncertainty of 0.50% (1σ) and a $^{238}U/^{206}Pb^*$ calibration uncertainty of 0.25% (1σ). Calibration uncertainties are included in the errors of $^{238}U/^{206}Pb^*$ ratios and dates listed in Table 1. Common-Pb corrections were applied to all analyses using contemporaneous isotopic compositions determined according to the model of Stacey and Kramers (1975).

Results

A total of 117 analyses were obtained from 116 zircons. Results are listed in Table 1, and shown in a concordia diagram (Fig. 3) and a probability diagram (Fig. 4).

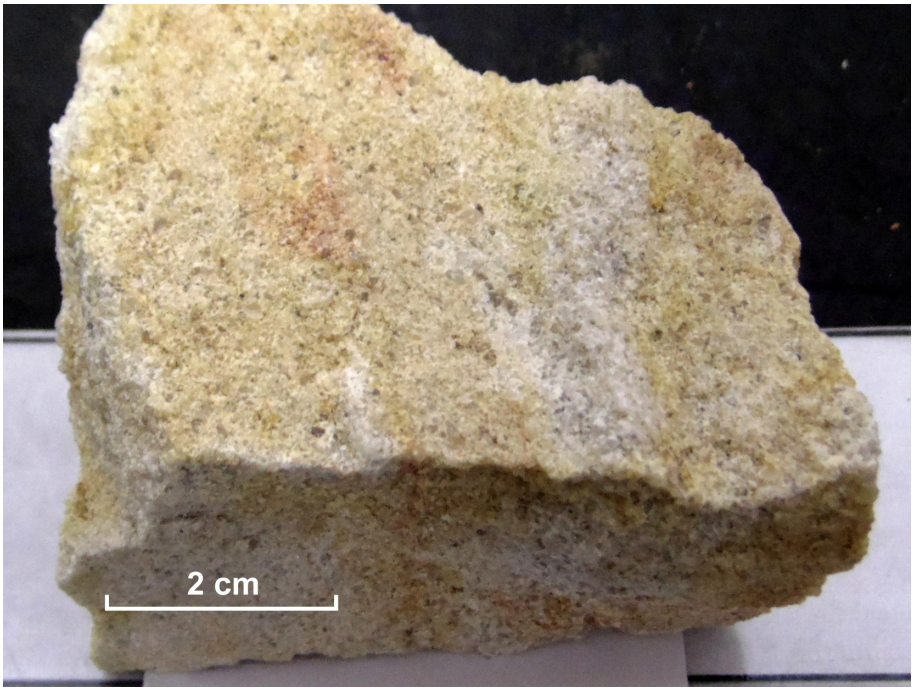

Figure 1. Hand specimen image of sample 215316: metasandstone, Midlands Road

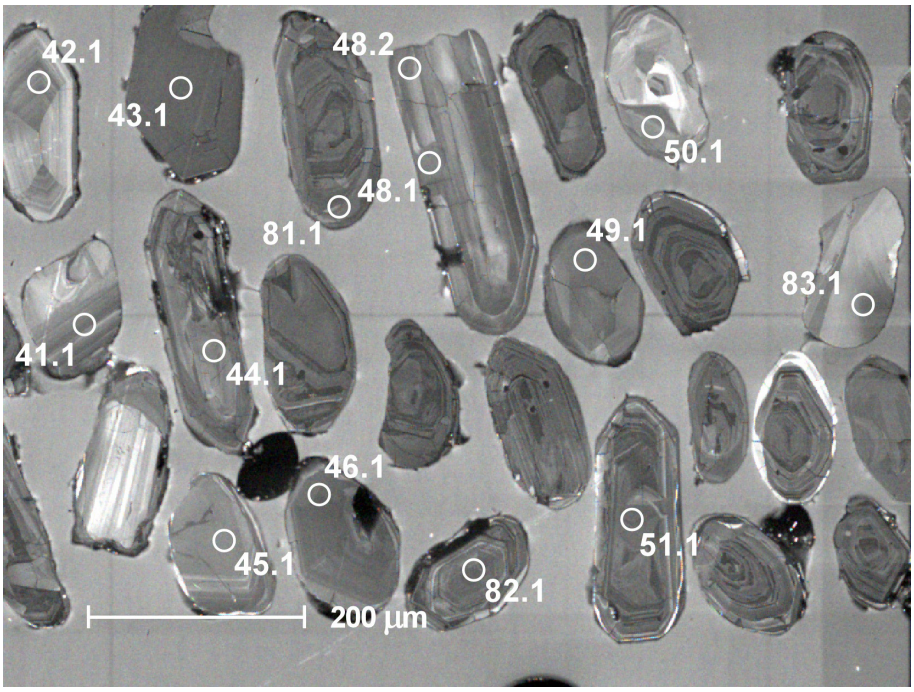

Figure 2. Cathodoluminescence image of representative zircons from sample 215316: metasandstone, Midlands Road. Numbered circles indicate the approximate locations of analysis sites

Table 1. Ion microprobe analytical results for zircons from sample 215316: metasandstone, Midlands Road

Group ID	Spot no.	Grain. spot	^{238}U (ppm)	^{232}Th (ppm)	$^{232}Th/^{238}U$	f204 (%)	$^{238}U/^{206}Pb$	± 1σ	$^{207}Pb/^{206}Pb$	± 1σ	$^{238}U/^{206}Pb^*$	± 1σ	$^{207}Pb^*/^{206}Pb^*$	± 1σ	$^{238}U/^{206}Pb^*$ date (Ma)	±1σ	$^{207}Pb^*/^{206}Pb^*$ date (Ma)	±1σ	Disc. (%)
Y	50	48.2	214	39	0.18	0.181	3.093	0.043	0.10955	0.00064	3.093	0.043	0.10955	0.00064	1806	22	1792	11	-0.8
Y	49	48.1	198	40	0.20	0.202	3.139	0.043	0.11074	0.00065	3.139	0.043	0.11074	0.00065	1783	22	1812	11	1.6
S	68	66.1	287	63	0.22	0.005	2.015	0.027	0.18039	0.00060	2.015	0.027	0.18034	0.00060	2598	29	2656	6	2.2
S	14	14.1	150	66	0.44	0.018	1.980	0.029	0.18150	0.00082	1.981	0.029	0.18134	0.00082	2635	32	2665	8	1.1
S	44	43.1	330	151	0.46	-0.008	1.953	0.026	0.18146	0.00055	1.953	0.026	0.18153	0.00055	2666	29	2667	5	0.1
S	93	91.1	105	44	0.42	-0.033	1.984	0.029	0.18178	0.00156	1.983	0.029	0.18207	0.00159	2632	32	2672	14	1.5
S	42	41.1	161	99	0.61	0.000	1.993	0.028	0.18250	0.00077	1.993	0.028	0.18250	0.00077	2621	31	2676	7	2.0
S	52	50.1	104	60	0.58	0.026	1.937	0.038	0.18525	0.00099	1.937	0.038	0.18501	0.00100	2683	43	2698	9	0.6
S	114	112.1	324	149	0.46	-0.022	1.985	0.020	0.18625	0.00092	1.984	0.020	0.18644	0.00093	2631	22	2711	8	3.0
S	13	13.1	131	22	0.17	-0.046	1.885	0.018	0.18656	0.00094	1.884	0.018	0.18697	0.00096	2745	22	2716	8	-1.1
S	7	7.1	285	98	0.34	0.018	1.933	0.026	0.18964	0.00123	1.933	0.026	0.18948	0.00123	2688	30	2738	11	1.8
S	18	18.1	249	48	0.19	0.000	1.964	0.015	0.18983	0.00065	1.964	0.015	0.18983	0.00065	2653	18	2741	6	3.2
S	35	34.1	287	199	0.69	0.017	1.916	0.031	0.19012	0.00059	1.916	0.031	0.18997	0.00059	2707	36	2742	5	1.3
S	77	75.1	272	95	0.35	-0.005	1.937	0.026	0.19031	0.00063	1.937	0.026	0.19035	0.00063	2683	30	2745	5	2.3
S	96	94.1	130	96	0.74	0.104	1.915	0.026	0.19160	0.00144	1.917	0.026	0.19067	0.00151	2706	30	2748	13	1.5
S	51	49.1	151	95	0.63	0.042	1.904	0.027	0.19160	0.00081	1.905	0.027	0.19123	0.00083	2720	32	2753	7	1.2
S	54	52.1	206	190	0.92	0.006	1.911	0.036	0.19194	0.00072	1.911	0.036	0.19189	0.00072	2713	42	2758	6	1.6
S	37	36.1	131	61	0.47	0.019	1.944	0.028	0.19406	0.00088	1.945	0.028	0.19389	0.00089	2675	32	2775	7	3.6
S	108	106.1	159	100	0.63	0.000	1.900	0.024	0.19421	0.00136	1.900	0.024	0.19421	0.00136	2726	29	2778	11	1.9
S	32	31.1	108	50	0.46	-0.034	1.864	0.027	0.19608	0.00096	1.863	0.027	0.19638	0.00098	2769	34	2796	8	1.0
S	67	65.1	193	63	0.32	0.028	1.910	0.031	0.19898	0.00077	1.911	0.031	0.19873	0.00078	2714	36	2816	6	3.6
S	16	16.1	81	31	0.38	0.052	1.651	0.018	0.23142	0.00367	1.652	0.018	0.23095	0.00368	3051	27	3059	25	0.2
S	2	2.1	120	61	0.51	0.104	1.557	0.016	0.25445	0.00113	1.558	0.016	0.25354	0.00116	3196	27	3207	7	0.3
S	111	109.1	151	97	0.64	-0.017	1.602	0.021	0.25375	0.00149	1.601	0.021	0.25391	0.00150	3128	32	3209	9	2.5
S	15	15.1	157	151	0.96	0.025	1.606	0.032	0.25529	0.00091	1.607	0.032	0.25507	0.00092	3119	50	3217	6	3.0
S	88	86.1	324	153	0.47	0.056	1.587	0.016	0.25626	0.00227	1.588	0.016	0.25578	0.00228	3149	25	3221	14	2.2
S	5	5.1	109	106	0.97	0.000	1.545	0.015	0.25588	0.00107	1.545	0.015	0.25588	0.00107	3218	26	3222	7	0.1
S	80	78.1	94	121	1.28	0.055	1.521	0.024	0.25718	0.00190	1.522	0.024	0.25670	0.00194	3256	40	3227	12	-0.9
S	85	83.1	79	75	0.94	0.094	1.539	0.025	0.25769	0.00203	1.540	0.025	0.25687	0.00208	3226	42	3228	13	0.1
S	87	85.1	44	35	0.79	-0.058	1.580	0.034	0.25646	0.00276	1.579	0.034	0.25697	0.00280	3162	54	3228	17	2.0

Table 1. continued

Group ID	Spot no.	Grain. spot	^{238}U (ppm)	^{232}Th (ppm)	$^{232}Th/^{238}U$	f204 (%)	$^{238}U/^{206}Pb$ ± 1σ		$^{207}Pb/^{206}Pb$ ± 1σ		$^{238}U/^{206}Pb^*$ ± 1σ		$^{207}Pb^*/^{206}Pb^*$ ± 1σ		$^{238}U/^{206}Pb^*$ date (Ma) ±1σ		$^{207}Pb^*/^{206}Pb^*$ date (Ma) ±1σ		Disc. (%)
S	17	17.1	69	79	1.15	0.099	1.570	0.018	0.25816	0.00137	1.571	0.018	0.25730	0.00141	3175	30	3230	9	1.7
S	79	77.1	245	240	0.98	0.099	1.604	0.017	0.25868	0.00122	1.605	0.017	0.25782	0.00125	3122	27	3233	8	3.5
S	89	87.1	165	96	0.58	-0.030	1.555	0.019	0.25766	0.00142	1.555	0.019	0.25792	0.00143	3201	32	3234	9	1.0
S	28	27.1	223	44	0.20	0.000	1.562	0.021	0.25849	0.00080	1.562	0.021	0.25849	0.00080	3191	35	3238	5	1.4
S	104	102.1	232	31	0.13	-0.022	1.566	0.017	0.25850	0.00122	1.566	0.017	0.25869	0.00123	3183	28	3239	7	1.7
S	47	46.1	180	160	0.89	0.000	1.566	0.022	0.25873	0.00082	1.566	0.022	0.25873	0.00082	3184	35	3239	5	1.7
S	9	9.1	176	113	0.64	0.025	1.450	0.013	0.25902	0.00092	1.450	0.013	0.25880	0.00093	3381	24	3239	6	-4.4
S	90	88.1	162	103	0.63	0.016	1.560	0.020	0.25894	0.00148	1.560	0.020	0.25880	0.00149	3193	32	3239	9	1.4
S	3	3.1	137	96	0.70	0.014	1.611	0.015	0.25915	0.00169	1.611	0.015	0.25903	0.00169	3113	23	3241	10	3.9
S	76	74.1	239	144	0.60	0.016	1.584	0.025	0.25919	0.00074	1.584	0.025	0.25905	0.00074	3155	40	3241	5	2.7
S	64	62.1	162	143	0.89	0.012	1.610	0.029	0.25945	0.00090	1.610	0.029	0.25935	0.00090	3115	44	3243	5	3.9
S	10	10.1	120	114	0.95	-0.016	1.602	0.024	0.25928	0.00104	1.602	0.024	0.25942	0.00104	3127	38	3243	6	3.6
S	95	93.1	191	115	0.60	-0.028	1.532	0.018	0.25930	0.00137	1.531	0.018	0.25955	0.00138	3240	31	3244	8	0.1
S	107	105.1	85	70	0.82	0.031	1.563	0.026	0.26033	0.00204	1.563	0.026	0.26006	0.00205	3188	42	3247	12	1.8
S	100	98.1	289	88	0.30	-0.017	1.620	0.017	0.26015	0.00109	1.619	0.017	0.26030	0.00110	3100	26	3249	7	4.6
S	23	22.1	312	133	0.43	0.017	1.608	0.021	0.26094	0.00068	1.608	0.021	0.26080	0.00068	3117	34	3252	4	4.1
S	55	53.1	268	175	0.65	-0.011	1.526	0.021	0.26091	0.00072	1.526	0.021	0.26100	0.00072	3249	35	3253	4	0.1
S	101	99.1	79	27	0.35	0.092	1.572	0.026	0.26243	0.00204	1.574	0.026	0.26163	0.00210	3171	41	3257	13	2.6
S	117	115.1	208	57	0.27	0.025	1.579	0.018	0.26252	0.00132	1.579	0.018	0.26230	0.00133	3163	30	3261	8	3.0
S	39	38.1	108	114	1.06	-0.020	1.577	0.024	0.26242	0.00120	1.577	0.024	0.26259	0.00120	3166	38	3262	7	3.0
S	46	45.1	109	54	0.50	0.065	1.549	0.023	0.26320	0.00105	1.550	0.023	0.26263	0.00107	3209	38	3263	6	1.7
S	97	95.1	194	43	0.22	-0.050	1.523	0.018	0.26266	0.00132	1.522	0.018	0.26310	0.00134	3255	30	3265	8	0.3
S	84	82.1	321	160	0.50	0.045	1.533	0.015	0.26488	0.00169	1.534	0.015	0.26449	0.00170	3236	26	3274	10	1.1
S	105	103.1	135	20	0.15	0.083	1.597	0.022	0.26556	0.00170	1.598	0.022	0.26483	0.00174	3133	34	3276	10	4.4
S	25	24.1	186	58	0.31	0.028	1.541	0.027	0.26618	0.00081	1.541	0.027	0.26594	0.00082	3224	45	3282	5	1.8
S	113	111.1	158	64	0.40	0.032	1.575	0.020	0.26624	0.00151	1.575	0.020	0.26596	0.00152	3169	32	3282	9	3.5
S	74	72.1	90	28	0.31	0.115	1.554	0.024	0.26859	0.00118	1.556	0.024	0.26759	0.00122	3200	39	3292	7	2.8
S	56	54.1	259	114	0.44	0.030	1.548	0.021	0.26789	0.00137	1.549	0.021	0.26762	0.00137	3211	35	3292	8	2.5
S	99	97.1	177	121	0.68	0.013	1.525	0.018	0.26782	0.00139	1.525	0.018	0.26771	0.00140	3250	31	3293	8	1.3
S	78	116.1	57	50	0.87	-0.140	1.530	0.030	0.26661	0.00256	1.528	0.030	0.26782	0.00265	3245	50	3293	16	1.5
S	20	20.1	184	109	0.60	0.010	1.592	0.032	0.26813	0.00250	1.592	0.032	0.26804	0.00250	3142	50	3295	15	4.6

Table 1. continued

Group ID	Spot no.	Grain. spot	^{238}U (ppm)	^{232}Th (ppm)	$^{232}Th/^{238}U$	f204 (%)	$^{238}U/^{206}Pb$ ± 1σ		$^{207}Pb/^{206}Pb$ ± 1σ		$^{238}U/^{206}Pb^*$ ± 1σ		$^{207}Pb^*/^{206}Pb^*$ ± 1σ		$^{238}U/^{206}Pb^*$ date (Ma) ±1σ		$^{207}Pb^*/^{206}Pb^*$ date (Ma) ±1σ		Disc. (%)
S	45	44.1	155	53	0.34	0.018	1.592	0.022	0.26826	0.00091	1.593	0.022	0.26811	0.00092	3141	36	3295	5	4.7
S	102	100.1	34	14	0.41	0.369	1.524	0.038	0.27154	0.00324	1.529	0.038	0.26833	0.00356	3243	63	3296	21	1.6
S	43	42.1	142	68	0.48	0.012	1.511	0.022	0.26871	0.00094	1.511	0.022	0.26861	0.00095	3273	37	3298	6	0.7
S	41	40.1	33	19	0.58	-0.025	1.564	0.029	0.26900	0.00189	1.564	0.029	0.26922	0.00190	3187	47	3302	11	3.5
S	21	21.1	103	45	0.44	0.066	1.499	0.015	0.26985	0.00109	1.500	0.015	0.26927	0.00111	3293	26	3302	6	0.3
S	75	73.1	94	80	0.86	0.021	1.518	0.034	0.26975	0.00121	1.518	0.034	0.26958	0.00122	3262	57	3304	7	1.3
S	22	76.1	122	122	1.00	0.015	1.508	0.027	0.26974	0.00105	1.509	0.027	0.26961	0.00106	3278	47	3304	6	0.8
S	34	33.1	149	78	0.52	-0.012	1.523	0.022	0.26969	0.00095	1.523	0.022	0.26979	0.00095	3254	37	3305	6	1.5
S	36	35.1	222	131	0.59	0.004	1.564	0.025	0.27022	0.00079	1.564	0.025	0.27019	0.00079	3187	41	3307	5	3.6
S	29	28.1	120	39	0.33	0.050	1.516	0.026	0.27132	0.00102	1.517	0.026	0.27088	0.00103	3264	45	3311	6	1.4
S	81	79.1	283	140	0.50	0.009	1.518	0.016	0.27135	0.00115	1.519	0.016	0.27128	0.00115	3262	27	3313	7	1.6
S	92	90.1	244	135	0.55	0.021	1.577	0.017	0.27145	0.00125	1.578	0.017	0.27127	0.00125	3165	28	3313	7	4.5
S	112	110.1	125	31	0.25	-0.019	1.550	0.033	0.27133	0.00168	1.550	0.033	0.27150	0.00169	3209	54	3315	10	3.2
S	12	12.1	117	54	0.46	0.035	1.464	0.015	0.27200	0.00113	1.464	0.015	0.27169	0.00114	3356	27	3316	7	-1.2
S	98	96.1	254	124	0.49	0.058	1.495	0.016	0.27261	0.00121	1.496	0.016	0.27210	0.00123	3300	29	3318	7	0.6
S	86	84.1	178	87	0.49	0.000	1.514	0.018	0.27219	0.00139	1.514	0.018	0.27219	0.00139	3269	31	3319	8	1.5
S	24	23.1	99	46	0.46	0.010	1.489	0.035	0.27298	0.00119	1.489	0.035	0.27289	0.00120	3312	60	3323	7	0.3
S	106	104.1	180	94	0.52	0.064	1.481	0.018	0.28583	0.00144	1.482	0.018	0.28529	0.00146	3325	32	3392	8	2.0
S	109	107.1	181	57	0.32	0.041	1.528	0.019	0.28841	0.00150	1.528	0.019	0.28805	0.00151	3245	32	3407	8	4.7
S	30	29.1	99	117	1.18	0.016	1.461	0.022	0.28916	0.00115	1.461	0.022	0.28902	0.00115	3362	40	3412	6	1.5
S	62	60.1	109	108	1.00	0.008	1.506	0.022	0.29027	0.00114	1.506	0.022	0.29020	0.00114	3282	39	3419	6	4.0
S	115	113.1	77	81	1.05	0.118	1.476	0.025	0.29187	0.00221	1.477	0.025	0.29085	0.00227	3333	44	3422	12	2.6
S	94	92.1	106	124	1.17	0.062	1.470	0.021	0.29318	0.00311	1.471	0.021	0.29265	0.00313	3344	38	3432	17	2.6
S	31	30.1	158	104	0.66	0.006	1.444	0.021	0.29280	0.00101	1.444	0.021	0.29274	0.00101	3392	38	3432	5	1.2
S	103	101.1	157	77	0.49	0.072	1.395	0.018	0.30926	0.00164	1.396	0.018	0.30865	0.00166	3483	35	3514	8	0.9
S	63	61.1	77	37	0.49	-0.029	1.394	0.040	0.32300	0.00403	1.393	0.040	0.32324	0.00404	3488	77	3585	19	2.7
D	66	64.1	307	141	0.46	-0.008	2.077	0.027	0.18255	0.00054	2.076	0.027	0.18262	0.00054	2534	28	2677	5	5.3
D	83	81.1	347	91	0.26	0.131	2.169	0.021	0.18561	0.00155	2.172	0.021	0.18443	0.00159	2442	20	2693	14	9.3
D	110	108.1	138	73	0.53	0.229	2.239	0.030	0.19261	0.00152	2.244	0.030	0.19057	0.00169	2376	27	2747	15	13.5

Table 1. continued

Group ID	Spot no.	Grain. spot	^{238}U (ppm)	^{232}Th (ppm)	$^{232}Th/^{238}U$	f204 (%)	$^{238}U/^{206}Pb$ ± 1σ		$^{207}Pb/^{206}Pb$ ± 1σ		$^{238}U/^{206}Pb^*$ ± 1σ		$^{207}Pb^*/^{206}Pb^*$ ± 1σ		$^{238}U/^{206}Pb^*$ date (Ma) ±1σ		$^{207}Pb^*/^{206}Pb^*$ date (Ma) ±1σ		Disc. (%)
D	73	71.1	88	61	0.69	0.031	2.168	0.033	0.21162	0.00122	2.168	0.033	0.21135	0.00123	2445	31	2916	9	16.2
D	4	4.1	399	178	0.45	0.035	2.756	0.020	0.21393	0.00186	2.757	0.020	0.21362	0.00187	1995	13	2933	14	32.0
D	1	1.1	393	381	0.97	0.071	1.966	0.021	0.22466	0.00054	1.967	0.021	0.22403	0.00056	2650	24	3010	4	12.0
D	27	26.1	175	57	0.32	0.135	2.457	0.034	0.24992	0.00097	2.460	0.034	0.24874	0.00101	2199	26	3177	6	30.8
D	72	70.1	99	41	0.41	0.030	1.745	0.026	0.25023	0.00112	1.746	0.026	0.24996	0.00113	2919	36	3185	7	8.3
D	53	51.1	306	32	0.10	0.010	1.711	0.025	0.25195	0.00066	1.711	0.025	0.25187	0.00066	2967	36	3197	4	7.2
D	65	63.1	289	137	0.48	0.041	1.746	0.043	0.25371	0.00070	1.747	0.043	0.25336	0.00071	2917	57	3206	4	9.0
D	6	6.1	188	79	0.42	0.035	1.692	0.032	0.25394	0.00082	1.693	0.032	0.25363	0.00083	2992	45	3208	5	6.7
D	71	69.1	156	40	0.25	0.013	1.696	0.031	0.25422	0.00094	1.696	0.031	0.25410	0.00095	2987	44	3210	6	6.9
D	38	37.1	212	164	0.77	0.006	1.746	0.024	0.25486	0.00087	1.746	0.024	0.25481	0.00087	2919	33	3215	5	9.2
D	11	11.1	91	83	0.91	0.084	1.698	0.018	0.25697	0.00126	1.699	0.018	0.25624	0.00129	2983	26	3224	8	7.5
D	61	59.1	197	69	0.35	0.000	1.706	0.036	0.25667	0.00081	1.706	0.036	0.25667	0.00081	2974	50	3226	5	7.8
D	57	55.1	217	126	0.58	0.034	1.648	0.023	0.25814	0.00081	1.649	0.023	0.25784	0.00082	3056	34	3234	5	5.5
D	82	80.1	186	148	0.80	0.031	1.782	0.021	0.25834	0.00144	1.783	0.021	0.25807	0.00145	2871	28	3235	9	11.3
D	8	8.1	107	54	0.50	0.093	1.693	0.017	0.25923	0.00118	1.694	0.017	0.25842	0.00121	2991	24	3237	7	7.6
D	116	114.1	295	147	0.50	0.080	1.898	0.019	0.26033	0.00117	1.900	0.020	0.25963	0.00120	2726	23	3245	7	16.0
D	91	89.1	91	91	1.01	0.133	1.663	0.026	0.26155	0.00189	1.666	0.026	0.26039	0.00197	3032	38	3249	12	6.7
D	40	39.1	184	57	0.31	0.030	1.685	0.024	0.26449	0.00200	1.686	0.024	0.26423	0.00201	3003	35	3272	12	8.2
D	58	56.1	175	164	0.94	0.011	1.759	0.042	0.26464	0.00224	1.759	0.042	0.26454	0.00224	2902	56	3274	13	11.4
D	70	68.1	242	92	0.38	0.039	1.632	0.033	0.26513	0.00085	1.633	0.033	0.26480	0.00086	3080	49	3275	5	6.0
D	19	19.1	105	59	0.56	0.029	1.653	0.017	0.26534	0.00115	1.653	0.017	0.26509	0.00116	3049	25	3277	7	7.0
D	48	47.1	236	137	0.58	0.044	1.760	0.024	0.26647	0.00084	1.761	0.024	0.26609	0.00085	2899	32	3283	5	11.7
D	26	25.1	369	125	0.34	0.018	1.620	0.031	0.26727	0.00062	1.620	0.031	0.26711	0.00062	3099	47	3289	4	5.8
D	69	67.1	143	66	0.47	0.066	1.698	0.024	0.26969	0.00110	1.699	0.024	0.26912	0.00112	2984	35	3301	7	9.6
D	60	58.1	145	33	0.23	0.042	1.731	0.031	0.27350	0.00101	1.732	0.031	0.27314	0.00102	2938	42	3324	6	11.6
D	33	32.1	146	180	1.23	0.091	1.604	0.024	0.28682	0.00103	1.606	0.024	0.28603	0.00105	3121	37	3396	6	8.1
D	59	57.1	149	106	0.71	0.062	1.554	0.027	0.29065	0.00358	1.555	0.027	0.29012	0.00359	3201	44	3418	19	6.4

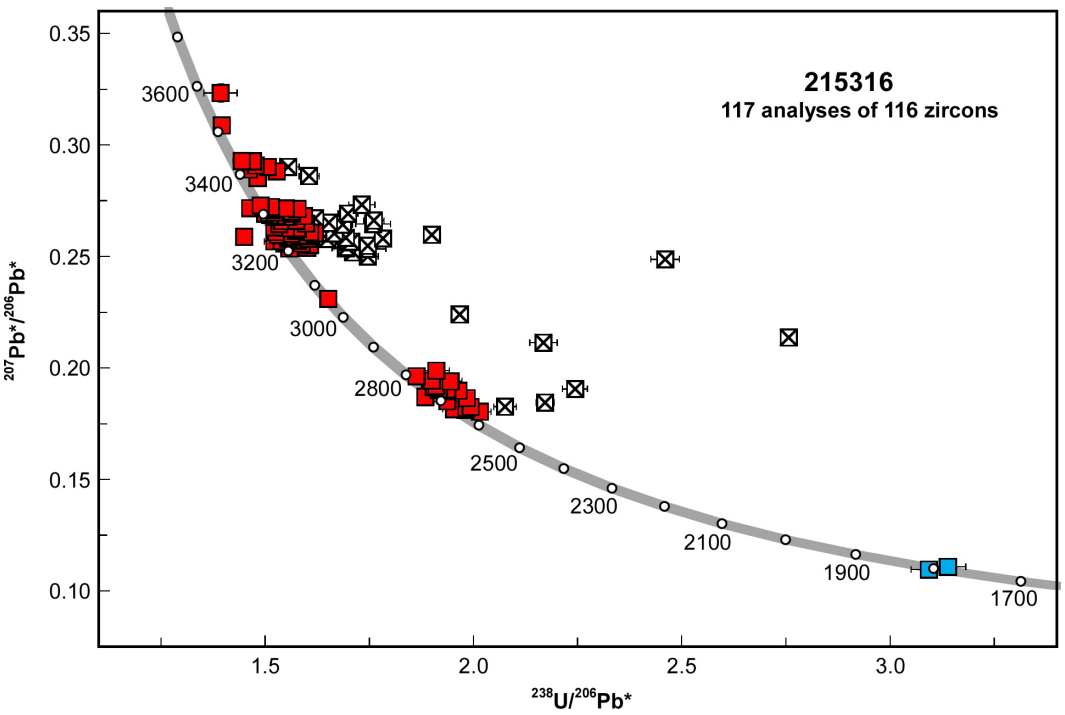

Figure 3. U–Pb analytical data for zircons from sample 215316: metasandstone, Midlands Road. Blue squares indicate Group Y (youngest detrital zircon); red squares indicate Group S (older detrital zircons); crossed squares indicate Group D (discordance >5%)

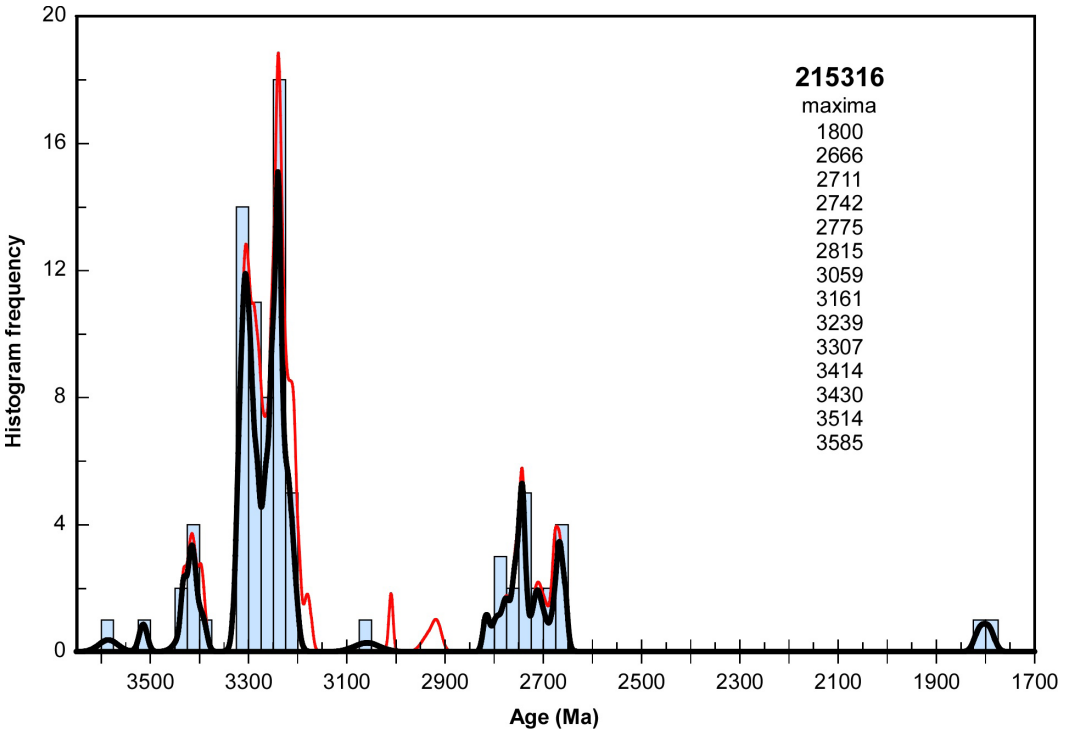

Figure 4. U–Pb analytical data for zircons from sample 215316: metasandstone, Midlands Road. Thick curve, maxima values, and frequency histogram (bin width 25 Ma) include data with discordance <5% (87 analyses of 86 zircons); thin curve includes all data (117 analyses of 116 zircons)

Interpretation

The analyses are concordant to strongly discordant (Fig. 3). Thirty analyses are >5% discordant. The dates obtained from these 30 analyses (Group D; Table 1) are unreliable, and are considered not to be geologically significant. The remaining 87 analyses can be divided into two groups, based on their $^{207}Pb^*/^{206}Pb^*$ ratios.

Group Y comprises two analyses of one zircon (Table 1), which yield a concordia age of 1800 ± 13 Ma (2σ, MSWD = 0.82).

Group S comprises 85 analyses of 85 zircons (Table 1), which yield $^{207}Pb^*/^{206}Pb^*$ dates of 3585–2656 Ma.

It is possible that all of the analyses in Group Y and S are of unmodified detrital zircons, in which case the date of 1800 ± 13 Ma for the two analyses in Group Y represents a maximum age of deposition for the sedimentary precursor. A more conservative estimate of the maximum depositional age can be based on the weighted mean $^{207}Pb^*/^{206}Pb^*$ date of 2665 ± 9 Ma (MSWD = 1.3) for the youngest five analyses in Group S.

The data for combined Groups Y and S indicate significant age components at c. 3430, 3414, 3307, 3239, 2742, 2711, and 2666 Ma, based on contributions from approximately 4, 5, 19, 20, 6, 3, and 5 analyses, respectively. These are interpreted as the ages of zircon-crystallizing rocks in the detrital source region(s), or the ages of detrital components within sediments which have been reworked into this rock.

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